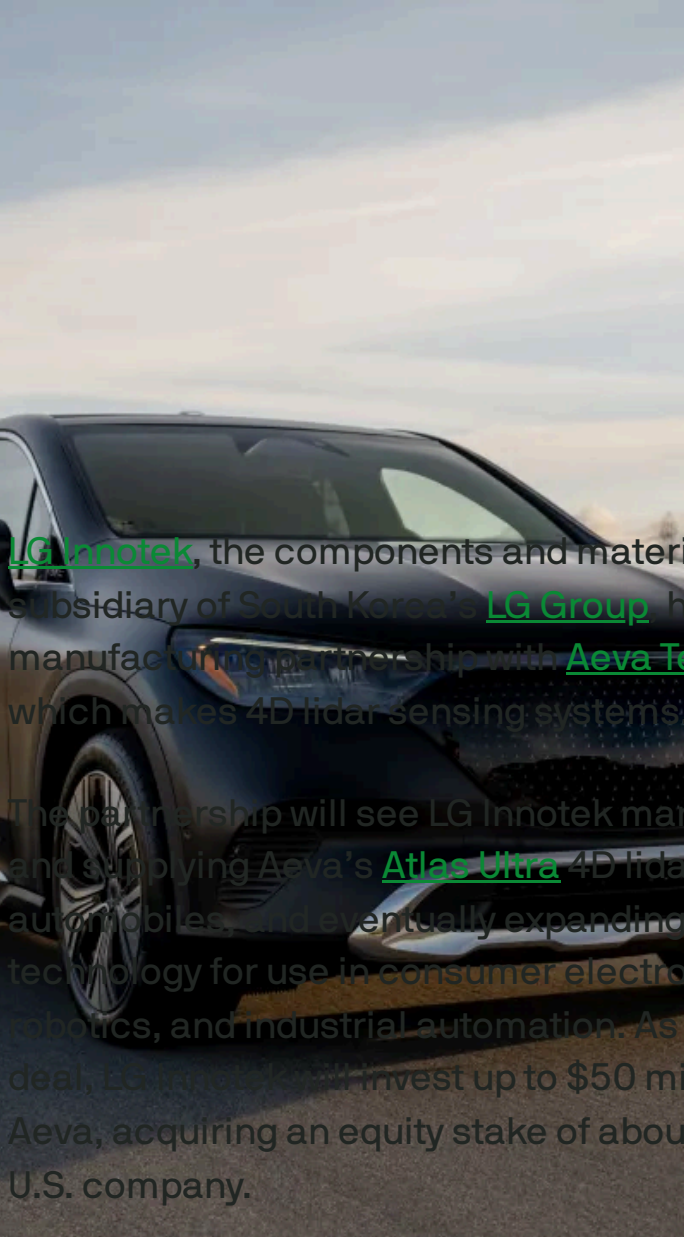




TRANSPORTATION



Aeva strikes lidar chip-making deal with LG subsidiary



[LG Innotek](#), the components and materials subsidiary of South Korea's [LG Group](#), has struck a manufacturing partnership with [Aeva Technologies](#), which makes 4D lidar sensing systems.

The partnership will see LG Innotek manufacturing and supplying Aeva's [Atlas Ultra](#) 4D lidar sensor for automobiles, and eventually expanding the technology for use in consumer electronics, robotics, and industrial automation. As part of the deal, LG Innotek will invest up to \$50 million in Aeva, acquiring an equity stake of about 6% in the U.S. company.

“The first part of the partnership is focused on the automotive sector,” Aeva’s co-founder and CEO Soroush Salehian told TechCrunch. “LG Innotek will act as our manufacturing partner for some of the top 10 global passenger vehicle OEMs.”

Founded in 2017, Aeva says it has poured nearly half a billion dollars into developing what it calls “4D lidar on a chip.” Unlike conventional lidar systems that rely on time-of-flight measurements to estimate distance, Aeva’s frequency modulated continuous wave-based perception technology measures both distance and velocity for every pixel in real time, Salehian said.

“It’s like going from black-and-white to color camera. We get this velocity information as a new dimension,” he explained.

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Aeva says it has integrated the entire lidar system, including optics, into a silicon photonics module. This miniaturized, chip-scale design enables more efficient production and integration, particularly for scaling in markets such as automotive, robotics, and consumer electronics, Salehian said.

The company plans to put the capital from the deal toward product development and augmenting its team. “Over the past eight years, we’ve made [the hardware] super small, and we’re sticking to that trajectory. We’re going to make it down to a monolithic, single-chip,” Salehian said.

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Aeva is also targeting the manufacturing sector and has struck partnerships Nikon in Japan and Sick AG in Germany. The company is also expanding into smart infrastructure and transportation, providing security and monitoring solutions at major U.S. airports, including SFO and JFK Terminal One.



**Kate Park**  
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Kate Park is a reporter at TechCrunch, with a focus on technology, startups and venture capital in Asia. She...

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# Aeva snags VW investment with smaller, longer-range lidar

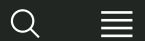
Kirsten Korosec — 1:45 AM PST · December 12, 2019

Lidar startup Aeva has deepened its relationship with VW Group with a new investment from Porsche Automobili Holding SE, thanks to a next-generation sensor that is headed for the ID Buzz AV, an electric reboot of the automaker's iconic bus that will be used as autonomous taxis.

Aeva's newest lidar product, called Aeries, has a 120-degree field-of-view — twice as much as its first product — and yet is half the size and uses less power. All of the components of the new lidar fit onto a single chip, an achievement that Aeva CEO



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Audi subsidiary Autonomous Intelligent Driving, or AID. The unit, which falls under the VW Group, is using Aeva lidar sensors in a fleet of autonomous electric e-trons that were being tested in Munich.

Aeva has developed what it describes as “4D lidar” that can measure distance as well as instant velocity without losing range, all while preventing interference from the sun or other sensors.

Lidar, or [light detection and ranging radar](#), measures distance. It's considered by many as a critical and necessary sensor for autonomous vehicles. Traditional lidar sensors are able to determine distance by sending out high-power

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pulses of light outside the visible spectrum and then tracking how long it takes for each of those pulses to return. As they come back, the direction of, and distance to, whatever those pulses hit are recorded as a point and eventually forms a 3D map.

The Aeries lidar sensor meets the final production requirements for autonomous driving robotaxis and large volume customers working on advanced driver assistance systems, and will be available for use in development vehicles in the first half of 2020, the company said.

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"It checks all the boxes and requirements in achieving high performance," said Alex Hitzinger, senior vice president of autonomous driving at VW Group and CEO of VW Autonomy, an autonomous development unit created earlier this year.

Specifically, Hitzinger pointed to the lidar sensor's high resolution, long range and small size.

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“Also, Aeva’s lidar measures the velocity for every point, which is a big deal for perception software and helps to significantly simplify the tasks perception like object classification for critical objects such as pedestrians at far distances,” Hitzinger said in an email to TechCrunch, adding that it’s the best solution on the market.

Volkswagen is planning to launch an ID Buzz AV for robotaxi applications in 2022. The vehicle will be the base platform for the development of the automaker’s self-driving system that will enable VW Autonomy to scale AV technology across the VW Group brands of vehicles afterwards.

Topics: [Aeva](#) [automotive industry](#) [AV](#) [driver](#) [Lidar](#)

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## Audi self-driving unit taps newcomer Aeva for its unique lidar

Kirsten Korosec — 11:10 AM PDT · April 17, 2019

Audi's self-driving unit has tapped a startup with a unique approach to lidar as it ramps up testing in Munich using a fleet of autonomous electric e-tron crossover vehicles.

Audi subsidiary [Autonomous Intelligent Driving](#), or AID, said Wednesday it's using lidar sensors developed by Aeva, a startup founded just two years ago by veterans of Apple and Nikon.

Aeva, a Mountain View, Calif.-based company started by Soroush Salehian and Mina Rezk, has developed what it describes as "4D lidar" that can measure distance as well as instant velocity

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without losing range, all while preventing interference from the sun or other sensors. Move past the 4D branding-speak, and the tech is compelling.

Lidar, or [light detection and ranging radar](#), measures distance. It's considered by many (with Tesla as one exception) in the emerging automated driving industry as a critical and necessary sensor. And for years, that industry has been dominated by Velodyne.

Today, there are dozens of lidar startups that have popped up with promises of technological breakthroughs that will offer lower-cost sensors with better resolution and accuracy than Velodyne. It's a promise that is fraught with challenges, notably the ability to scale up manufacturing.

Traditional lidar sensors are able to determine distance by sending out high-power pulses of light outside the visible spectrum and then tracking how long it takes for each of those pulses to return. As they come back, the direction of, and distance to, whatever those pulses hit are recorded as a point and eventually forms a 3D map.

Aeva's sensors emit a continuous low-power laser, which allows them to sense instant velocity of every point in the frame at ranges up to 300 meters, the company says. In other words, Aeva's sensors can determine distance and direction, as well as speed of the objects coming to or moving away from them.

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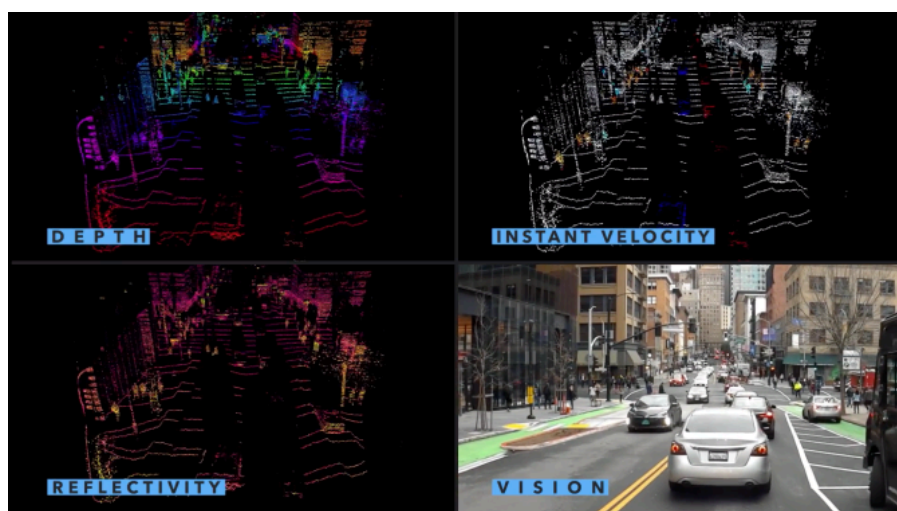
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This is a handy perception feature for autonomous vehicles operating in an environment of objects that travel at different speeds, like pedestrians, bicycles and vehicles.

Aeva, backed by investors including Lux Capital and Canaan Partners, says its sensors are also unique because they're "free" from interference from other sensors or sunlight.

It was this combination of long-range perception, instantaneous velocity measurements at cm/s precision and robustness to interferences that sold AID CTO Alexandre Haag on the Aeva sensors.

Aeva spent the past 18 months going through a validation process with Audi and parent company Volkswagen. This announcement confirms that Aeva has made it past a critical hurdle in Audi's AV plans. Aeva's sensors are already on Audi e-tron

development vehicles in Munich. The automaker plans to bring autonomous driving to urban mobility services within the next few years.

Interference is possible and can cause a stream of random points on a 3D map if the lidar is pointed directly at the sun or if there are multiple sensors on the same vehicle. Lidar companies have instituted various techniques to prevent interference patterns; autonomous vehicle developers also account for potential interference problems from the sun and snow by creating algorithms to reject these kinds of outliers.

Still, Salehian argues that interference is a significant challenge.

When you talk about the challenge of building to scale and designing for mass scale, it's not just about how easily they can be manufactured, Salehian contends. "It's also about having these things work in unison together on a row. So when you're talking about hundreds of thousands of these cars, that's a big deal."

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